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Sofia Sannino

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Professional Experience

Nestlé, R&D Manufacturing Intern — Bern, Switzerland — **July 2026 – present**

- Developing a forecasting model to predict chemical concentrations of volatiles linked to Maillard reaction in spray drying, aiming to optimize the process.

Politecnico di Milano, Students' Representative in the Board of Governors — Milan, Italy — **June 2023 – May 2025**

- Elected as one of two students' representatives on the University Board of Governors, with full voting rights and participation in key decision-making processes, including university budget approval and allocation of students' associations funds.
- Daily collaboration with the Rector, directors and administration to improve students' life on campus.
- Represented Politecnico di Milano students in the Municipal Roundtable on Student Housing in Milan.

Education

EPFL (École Polytechnique Fédérale de Lausanne), Master Degree in Computational Science and Engineering, Double Degree program — Lausanne, Switzerland — **Sep 2025 – present**

- Among the five master students selected to take part to the Double Degree program between EPFL and Politecnico di Milano.
- Average grade: 5.6 / 6

Politecnico di Milano, Master Degree in Mathematical Engineering - Computational Science and Computational Learning, Double Degree program — Milan, Italy — **Sep 2024 – present**

Alta Scuola Politecnica, International honour program between Politecnico di Milano and Torino — Milan, Italy — **Dec 2024 – Jan 2026**

- Among the 90 master students selected, I worked with A2A (Italian energy company) in a multidisciplinary team to design a sustainable electric-vehicle charging business model.

Politecnico di Milano, Bachelor Degree in Mathematical Engineering — Milan, Italy — **Sep 2021 – Sep 2024**

- Topics: Advanced Calculus and Algebra, Probability, Statistics, Numerical Methods, PDEs, Physics, Control Theory, Computer Science, HPC.
- Final grade: 110 / 110 cum laude, GPA: 29.05 / 30

Liceo Guglielmo Marconi, Scientific Diploma — Parma, Italy — **Sep 2016 – June 2021**

- Final grade: 100 / 100 cum laude

Skills

Machine Learning and Signal Processing: Multi-modal deep learning; Contrastive learning; State-space models; Volumetric rendering; Image processing; Multi-variate time-series analysis; Reinforcement learning.

Mathematical Modelling and Numerical Methods: Advanced Numerical Methods for PDEs (Multidimensional heat equation and advection-diffusion problems) ; FEM (formulation, stability, error estimates); CFD for laminar and turbolent flows with FVM; Monte Carlo simulations for molecular dynamics.

Tools: Python, C++, C, Matlab, R, Git, Databricks, ANSYS Fluent

Languages: Italian (Native), English (C1), Spanish (B2), French (B1)

Projects

Code and reports can be found at the GitHub link in the link-tree on top of the CV.

Probabilistic Spatio-Temporal Modeling of Cardiac Deformation
Swiss Data Science Center
Computer Vision Lab EPFL
Feb 2026 - Jun 2026

- Defined a new probabilistic state-space model, representing cardiac motion time-series data by combining Gaussian primitives with Markovian Gaussian Processes (GP): images are represented by sparse control-points, expression of a latent GP, whose distribution is found solving the SDE obtained from the state-space formulation of the GP; the image is then rendered using 3D Gaussian Splatting. Successfully implemented a new deterministic control-points representation to pretrain the first layer of the state-space model, through which the sparse points cloud is obtained.

Language-Vision Deep Learning Model for Cancer Segmentation
Gustave Roussy Cancer Center, Paris
Oct 2025- Jan 2026

- Designed a PyTorch vision-language lesion segmentation model on CT scans, fusing radiologist text embeddings (BERT) with image features and tissue priors to segment the referenced lesion across anatomical sites; improved Dice score by 50 % vs their baseline and owned data preprocessing, training, validation and reporting with clinical stakeholders.

Learning Matches for Mesh Deformation via REINFORCE
Computer Vision Lab EPFL
April 2026 - May 2026

- Co-designed a new mesh deformation pipeline in which points correspondences are learned as a policy that can predict matches based on input shapes: in this reinforcement learning formulation the state is the current source-target configuration, the action is a correspondence proposal, the transition is represented by deformation steps, and the reward is the final fit quality, using REINFORCE as optimization algorithm. We show improvements with respect to the standard Chamfer distance optimization pipeline, testing our method on a dataset of 2D shapes.

Can a Formula 1 Car Drive on the Ceiling?
Numerical Flow Simulation
EPFL
Oct 2025- Jan 2026

- Worked in a team to simulate the unsteady aerodynamic interaction between an inverted Formula 1 car geometry and the surrounding flow field. Main contributions: setup and execution of CFD simulations in ANSYS Fluent (mesh generation with FVM, turbulence-model selection, solver configuration). Successfully validated our results against the literature despite limited resources (yes, a Formula 1 car can drive on a ceiling).

Deterministic chaos
Turbulence
EPFL
April 2026- May 2026

- Study of the validity and limitations of turbulence Kolmogorov theory applied to real turbulent flows, using time series data of the down-stream velocity from a wind tunnel. The study highlights, through numerical simulations and data analysis, how the theory breaks due to intermittency phenomena. The presentation obtained a grade of 5.75/6.

2D Steady Diffusion-Convection Analysis with FVM
Numerical Flow Simulation
EPFL
Sep 2025- Nov 2025

- Designed a Matlab pipeline to analyse the temperature field in an internal flow by discretizing the governing partial differential equation with the Finite Volume Method, comparison of different discretization schemes and linear-system solvers, implementation and assessment of multiple boundary-condition formulations, solution verification and validation. Grade: 20.75/22.

What Causes University Dropout?
Statistical Inference Course
Politecnico di Milano
May 2024

- Designed a R pipeline to analyze Politecnico di Milano students data with regression models to understand which features influence dropout from university and academic performance. Grade 4/4.

Associations

Svoltastudenti, Member of the Leaders Team and Student Representative
2023 – 2025

- Led and won two university-wide electoral campaigns (>45,000 students), one as candidate and one as list representative.

- Coordinated a network of 100+ course representatives, ensuring information flow, policy alignment and operational support.
- Managed content creation on student representation topics for social media channels and the association's website.

Scholarships and Awards

Alta Scuola Politecnica Scholarship Founding

- Full tuition scholarship covering all university fees and and personal research fund to support project development within the program.

Politecnico di Milano Merit Scholarship for High GPA, A.Y. 2023/2024

- Full tuition covered for students with a GPA higher than 29/30.

Best Freshmen Award A.Y. 2021/2022

- €1,000 prize awarded to the top 2% of first-year students at Politecnico di Milano.

Licenses

TOEFL, 109/120 (English level C1), obtained in November 2024
Driving License B